

The printed geometric-mean conjecture already fails for scalars in arXiv:2401.00337

Candidate counterexample packet

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Status: candidate counterexample likely valid. This is a full negative answer to Conjecture 3.7 exactly as printed. It is also a semantic warning: the likely intended $t = \frac{1}{2}$ or $\sharp_t$ formulation is not settled here.

1 Source conjecture

The source defines the weighted geometric mean

$$A \sharp_t B = A^{1/2} (A^{-1/2} B A^{-1/2})^t A^{1/2}, \quad 0 \leq t \leq 1,$$

and writes $A \sharp B$ for the unweighted case $t = \frac{1}{2}$. Its final Conjecture 3.7 asks, for $1 \leq s \leq 2$, whether

$$\left\| \sum_{i=1}^m (A_i^s \sharp B_i^s)^r \right\| \leq \left\| \left[\left(\sum_i A_i \right)^{(1-t)srp/2} \left(\sum_i B_i \right)^{tsrp} \left(\sum_i A_i \right)^{(1-t)srp/2} \right]^{1/p} \right\| \quad (1)$$

for $p > 0$, $r \geq 1$, $rp \geq 1$, and every unitarily invariant norm [1]. The parameter t appears on the right but is neither introduced nor quantified in the conjecture; the left side contains $\sharp$, not $\sharp_t$.

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Finally, with an eye on (1.8) and Theorem 3.6, we state the following conjecture.

Conjecture 3.7. *Let $A_i, B_i \in \mathbb{P}_n$ for all $i = 1, \dots, m$. Let $1 \leq s \leq 2$. Then*

$$\left\| \sum_{i=1}^m (A_i^s \sharp B_i^s)^r \right\| \leq \left\| \left(\left(\sum_{i=1}^m A_i \right)^{\frac{(1-t)srp}{2}} \left(\sum_{i=1}^m B_i \right)^{tsrp} \left(\sum_{i=1}^m A_i \right)^{\frac{(1-t)srp}{2}} \right)^{\frac{1}{p}} \right\|,$$

for all $p > 0$, for all $r \geq 1$ such that $rp \geq 1$ and for all unitarily invariant norms.

Note that, using (1.8) and Theorem 3.6, the above conjecture is true for the cases $s = 1$ and $s = 2$, respectively. The cases $s \in (1, 2)$ remains open.

The complete conjecture and open-case sentence in the source PDF, page 12.

2 Proof Intuition

The displayed inequality is required for matrices of every size, so it must survive the one-dimensional case. There all noncommutativity disappears and the unweighted mean on the left contributes equal powers of A and B . The right side instead contributes the unequal powers $1 - t$ and t . Their ratio is a single power of A/B , which can be made larger than one unless $t = \frac{1}{2}$. The counterexample therefore targets the exact mismatch in the printed display; it does not rely on delicate matrix behavior.

3 Counterexample and exact scalar obstruction

Theorem 1. *Under the natural reading that $t \in [0, 1]$ is inherited from the preceding weighted-mean statements, (1) is false for every $s > 0$ as soon as $t \neq \frac{1}{2}$. In particular it is false in the stated open range $1 < s < 2$.*

Proof. Take $n = m = 1$, so positive definite matrices are positive scalars and every unitarily invariant norm is a positive multiple of the absolute value. Put $A_1 = A > 0$ and $B_1 = B > 0$. Because

$$A^s \sharp B^s = (A^s B^s)^{1/2} = (AB)^{s/2},$$

the two sides of (1) reduce, after cancelling the norm's constant factor, to

$$L = (AB)^{sr/2}, \quad R = A^{(1-t)sr} B^{tsr}.$$

Consequently

$$\frac{L}{R} = \left(\frac{A}{B} \right)^{sr(t-1/2)}.$$

If $t > \frac{1}{2}$, choose $A > B$; if $t < \frac{1}{2}$, choose $A < B$. In either case $L/R > 1$. Hence scalar validity for all positive A, B forces $t = \frac{1}{2}$.

For a concrete counterexample in the open range, choose

$$A = 4, \quad B = 1, \quad s = \frac{3}{2}, \quad r = p = 1, \quad t = 1.$$

All printed hypotheses are satisfied, while

$$L = (4 \cdot 1)^{3/4} = 2\sqrt{2} > 1 = 4^0 1^{3/2} = R.$$

□

The same computation shows that the printed endpoint assertions cannot hold for arbitrary t : the defect is independent of s . This internal inconsistency is further evidence that the final display lost either a subscript t on the left or the specialization $t = \frac{1}{2}$ on the right.

4 What remains open after repairing the statement

There are two coherent repairs:

1. replace $A_i^s \sharp B_i^s$ by $A_i^s \sharp_t B_i^s$ while retaining $t \in [0, 1]$ on the right; or
2. fix $t = \frac{1}{2}$ on the right and retain the unweighted geometric mean on the left.

For scalars, both repairs give equality. The source’s Theorem 3.6 is an unweighted result for $s \geq 2$, so the second repair is especially consistent with the sentence immediately preceding the conjecture. This packet makes no claim about either noncommutative repaired problem.

As a guard against mistaking a typographical counterexample for a substantive one, we ran a focused numerical upgrade search on the second repair. Random positive-definite 2×2 and 3×3 matrices (up to four summands), several values of $s \in (1, 2)$, r , p , Schatten norms, and Ky Fan norms gave 280,000 matrix configurations and 700,000 Ky Fan tests, with no violation; the largest ratio was 0.999999999798115. A differential-evolution maximization over four noncommuting real 2×2 positive-definite matrices, across 50 parameter regimes, likewise produced no ratio above 1 (the largest was 0.9999994503007108 among nontrivial reported runs). These computations are not a proof, but they support keeping the repaired conjecture open rather than claiming more than the scalar diagnosis establishes.

5 Verification and novelty scope

The accompanying SymPy script evaluates the displayed scalar example exactly and verifies $L - R = 2\sqrt{2} - 1 > 0$. No floating-point arithmetic is used in that verification. The two additional scripts reproduce the random and optimized checks via

```
conda run --no-capture-output -n sandbox python \
  code/search_repaired_conjecture.py
conda run --no-capture-output -n sandbox python \
  code/optimize_repaired_conjecture.py
```

A bounded search through 11 August 2026 covered the run’s registry, solution, attempt, and proof-gap indexes; the exact conjecture text; the arXiv identifier and title with *Conjecture 3.7*, *counterexample*, and *t-geometric mean*; and later papers surfaced by those searches. It found no later source explicitly correcting or resolving Conjecture 3.7. Novelty confidence is moderate pending specialist review. The literal scalar calculation is independent of novelty.

6 Conclusion

Conjecture 3.7 is false exactly as displayed: scalar validity already forces $t = \frac{1}{2}$, whereas its right side retains a free t and its left side is unweighted. The example $A = 4$, $B = 1$, $s = 3/2$, $r = p = t = 1$ is a full explicit counterexample. Human review should treat this as a semantic warning and decide which repaired noncommutative conjecture was intended.

References

- [1] S. Freewan and M. Hayajneh, *Norm inequalities involving geometric means*, arXiv:2401.00337 (2024). <https://arxiv.org/abs/2401.00337>.